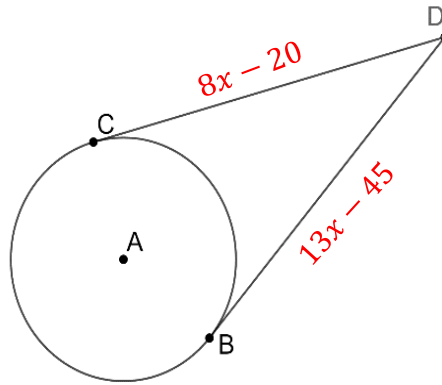


Geometry
Unit 12
Lesson 4 Practice

Name **Answer Key** _____
Date _____
Period _____

For questions 1-3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
 $CD = 8x - 20$ and $DB = 13x - 45$.



1. What is the value of x ?

$$\begin{aligned} 8x - 20 &= 13x - 45 \\ -5x &= -25 \\ \boxed{x = 5} \end{aligned}$$

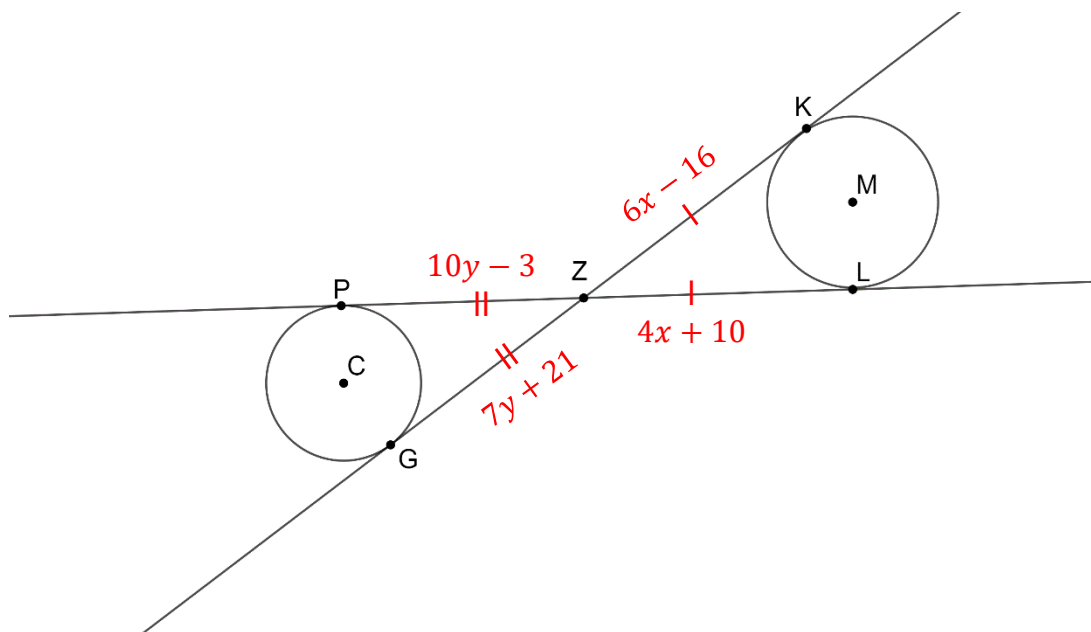
2. What is the length of $\overline{CD}$?

$$\begin{aligned} CD &= 8(5) - 20 \\ \boxed{CD = 20} \end{aligned}$$

3. What is the length of $\overline{DB}$?

$$\begin{aligned} DB &= 13(5) - 45 \\ 65 - 45 \\ \boxed{DB = 20} \end{aligned}$$

For questions 4-10, Circles C and M are shown. Lines $\overline{PL}$ and $\overline{GK}$ intersect at point Z . Line $\overline{PL}$ is tangent to circle C at point P and circle M at point L . Line $\overline{GK}$ is tangent to circle C at point G and circle M at point K . $PZ = 10y - 3$, $ZL = 4x + 10$, $GZ = 7y + 21$, $ZK = 6x - 16$.



4. Write an equation you can use to solve for x .

$$4x + 10 = 6x - 16$$

5. Solve the equation you wrote in question 4 for x .

$$10 = 2x - 16$$

$$26 = 2x$$

$$x = 13$$

6. Write an equation you can use to solve for y .

$$10y - 3 = 7y + 21$$

7. Solve the equation you wrote in question 6 for y .

$$3y - 3 = 21$$

$$3y = 24$$

$$y = 8$$

8. What is the length of $\overline{GK}$?

$$GK = GZ + ZK \quad (7 \cdot 8 + 21) + (6 \cdot 13 - 16)$$

$$77 + 62$$

$$GK = 139$$

9. What is the length of $\overline{GZ}$?

$$GZ = 7(8) + 21$$

$$GZ = 77$$

10. What is the length of $\overline{ZL}$?

$$ZL = 4(13) + 10$$

$$= 52 + 10$$

$$ZL = 62$$